The smallest n1(k) common to a triple

Problèmes ouverts — A. Dujella · 2026

Problem 5.13 (Dujella, 2026). For a positive integer k, denote by $n_1(k)$ the smallest (in the absolute value) non-zero integer for which there exists a triple $\{a, b, c\}$ of non-zero integers and a set N of integers, such that $|N| = k$, $n_1(k) \in N$, and $\{a, b, c\}$ is a $D(n)$-triple for all $n \in N$.

(a) Is it true that $n_1(5) = 36$?

(b) What can be said (assuming some plausible conjectures) about the asymptotic behaviour of $|n_1(k)|$ for large k ?

Remark. Sadowski (2026) answered question (a) showing that $n_1(5) \leqslant 4$. In fact, he showed that also $n_1(6) \leqslant 4$ with the following example : $\{a, b, c\} = \{30, 1056, 65520\}$, $N = \{4, 266436, 15248601, 21479364, 6393$ Furthermore, $n_1(7) \leqslant 36$ with $\{a, b, c\} = \{8, 1620, 257040\}$, $N = \{36, 857529, 1470564, 200322756, 238596849, 4520$